

Set-2.

1 Spanning Tree protocol (STP) Using packet Tracer

Aim:-

To design and configure a network with 2 switches connected redundancy using Spanning Tree protocol (STP) to prevent broadcast storms and network loops.

Procedure:-

1. Open Cisco packet Tracer
2. Add 2 switches (eg. 2960)
3. Add pcs and connect them to both service.
4. Connect the two switches using two links to create redundancy.
5. Configure IP address for the PCs.
6. STP is enabled by default on Cisco switches.
7. Check STP status using the CLI
8. Identify the Root bridge and blocked.
9. Test communication between PCs.

Configuration Steps

Switch 1:

enable

configure terminal

hostname sw1

spanning tree mode puvst

spanning tree vlan 1 root primary

end

spanning - tree

Switch 2 :

enable

configure terminal

hostname Sw2

spanning-tree mode pvlst

spanning-tree vlan 11 root secondary

end

show spanning tree

Result :-

STP has successfully configured the redundant link was placed in blocking state, preventing network loops and broadcast storm while maintaining network connectivity.

2) Network for 40 system in Two Buildings :-

Aim :-

To design and configure a network connecting 40 system across two building located 200 metres apart, using suitable transmission media and topology in Cisco packet tracer.

Procedure :-

1. Open Cisco packet tracer
2. Create Building 1 and Building 2
3. Add 20 pcs in each building
4. Add switches to connect the pcs
5. Link the building are 200m apart
6. Use fiber optic cable for inter building connection

↳ connect PC to switches using copper Ethernet using fiber.

↳ connect PCs to switches between the buildings using fiber.

8) Assign IP addresses to all PCs

9) Test Connectivity using the ping command

Configuration Steps :-

Building 1 :

192.168.1.1 - 192.168.1.20

192.168.1.21 - 192.168.1.40.

Building 2 :

192.168.1.21 - 192.168.1.40

Subnet mask :-

255.255.255.0

Example PC Configuration :-

IP Address : 192.168.1.1

Subnet mask : 255.255.255.0

Media :

→ PC → Switch : copper straight-through

→ Switch → Switch between building : fiber optic.

Result :-

A network connecting two systems in two buildings was successfully designed and configured - fiber - optic cable was used for the 200 m inter building connection, and communication between systems was successfully verified.

3. BOOTP packet analysis packets using Wireshark.

Aim :-

To capture and analyze BOOTP packets using Wireshark and study their header fields and payload.

Procedure :-

1. Install and open Wireshark.

2. Select the active network interface.

3. Start packet capture.

4. Generate network traffic that uses DHCP.

BOOTP.

5. Stop the capture.

6. Apply the filter.

bootp

7. Select a BOOTP / DHCP section.

8. Analyze the header fields and payload.

Configuration :-

No special configuration is required in Wireshark. Use the display filter :-
bootp.

Result :-

BOOTP packets were successfully captured and their header fields and payload were analyzed using Wireshark.

4. Class B Subnetting :-

Aim :-

To demonstrate class B subnetting for the network address 172.16.0.0 using the subnet mask 255.255.255.0 (1/24)

Procedure :-

1. Identify the given class B network address 172.16.0.0 using the subnet mask 255.255.255.0 (1/24).
2. Use the subnet mask 255.255.255.0
3. Calculate the number of subnets and hosts.
4. Identify the network, usable and broadcast addresses
5. Assign the required subnet address

Configuration :-

Network : 172.16.0.0/24

- Subnet mask. 255.255.255.0.
- number of /24 subnets from the original
/16 : 256
- usable hosts per subnet : 254.

Example :-

Network	usable hosts	Broad.
172.16.0.0	172.16.0.1 - 172.16.0.254	172.16
172.16.1.0	172.16.1.1 - 172.16.1.254	172.16
172.16.2.0	172.16.2.1 - 172.16.2.254	172.16

Result :-

The class B network 172.16.0.0 was successfully subnetted using the /24 subnet mask, provide 256 subnets with 254 usable hosts per subnet.